Digitally signed by WIN-IINGEN
Date: 2025.12.16 20:07:32 -06:00
Folio: NTCDS-CDMX-2025-31-23635
Normas Técnicas Complementarias para Diseño por Sismo - CDMX 2023
Location: MEX-CDMX

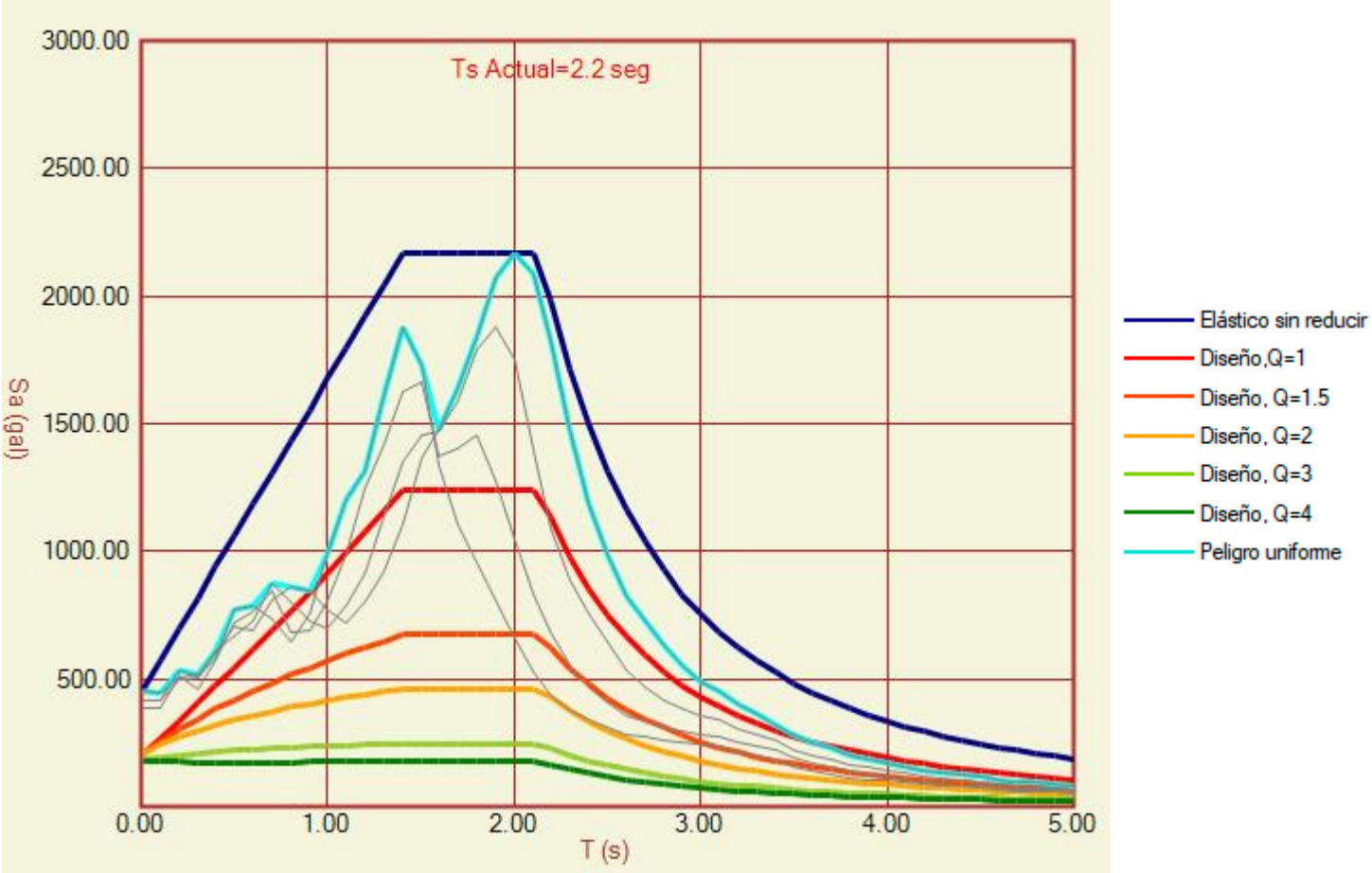
ESPECTRO ELÁSTICO DE DISEÑO

Solicitado por: Eduardo Marin
Correo electrónico: eduardo@earthquakeprotection.com

Folio: NTCDS-CDMX-2025-31-23635
Fecha de generación: 16/12/2025 20:07

Ubicación: Latitud: 19.424803803487205, Longitud: -99.15642805597997
Dirección del predio: Metro Cuahutemoc, Morelia 10

Factor de hiperestaticidad.k1	Nivel de diseño sismico	Amortiguamiento	Estructura de mampostería
1.00	Sismo infrecuente	0.050	No



a0 = 450 c = 2166 Ta = 1.4 Tb = 2.1 k = 0.355

Periodo	EPU	Elástico	ED(Q=1.0)	ED(Q=1.5)	ED(Q=2)	ED(Q=3)	ED(Q=4)
0.010	451.9	462.3	209.4	208.1	206.9	183.7	181.6
0.100	449.2	572.5	270.5	255.2	241.6	195.2	178.0
0.200	536.4	695.1	337.3	301.2	272.0	203.3	174.9
0.300	522.9	817.6	405.1	343.3	297.9	209.6	173.4
0.400	611.2	940.2	474.2	382.5	320.5	214.9	172.7
0.500	768.9	1062.7	544.7	419.1	340.5	219.6	172.5

0.600	789.1	1185.3	616.5	453.4	358.6	223.7	172.8
0.700	879.2	1307.8	689.6	485.8	375.0	227.5	173.2
0.800	858.9	1430.3	764.1	516.4	390.0	231.0	173.8
0.900	847.1	1552.9	839.8	545.5	404.0	234.3	174.6
1.000	984.5	1675.4	916.8	573.2	417.0	237.4	175.5
1.100	1205.2	1798.0	995.1	599.7	429.2	240.3	176.4
1.200	1313.0	1920.5	1074.7	625.1	440.7	243.2	177.4
1.300	1612.2	2043.1	1155.5	649.4	451.6	245.9	178.4
1.400	1875.5	2165.6	1237.5	672.8	462.0	248.5	179.4
1.500	1727.0	2165.6	1237.5	672.8	462.0	248.5	179.4
1.600	1480.1	2165.6	1237.5	672.8	462.0	248.5	179.4
1.700	1639.0	2165.6	1237.5	672.8	462.0	248.5	179.4
1.800	1843.1	2165.6	1237.5	672.8	462.0	248.5	179.4
1.900	2072.5	2165.6	1237.5	672.8	462.0	248.5	179.4
2.000	2165.6	2165.6	1237.5	672.8	462.0	248.5	179.4
2.100	2087.0	2165.6	1237.5	672.8	462.0	248.5	179.4
2.200	1826.8	1984.0	1133.7	621.2	427.8	230.7	166.7
2.300	1475.4	1717.6	981.5	544.5	376.8	204.0	147.7
2.400	1189.1	1498.7	856.4	480.5	334.0	181.5	131.6
2.500	990.7	1317.1	752.7	426.7	297.8	162.4	118.0
2.600	826.2	1165.3	665.9	381.1	266.9	146.1	106.3
2.700	731.4	1037.2	592.7	342.2	240.5	132.0	96.2
2.800	629.2	928.3	530.5	308.7	217.7	119.8	87.4
2.900	557.8	835.2	477.2	279.8	197.9	109.3	79.8
3.000	491.9	754.9	431.4	254.7	180.7	100.0	73.1
3.100	452.4	685.4	391.7	232.7	165.5	91.8	67.2
3.200	401.3	624.9	357.1	213.4	152.2	84.6	62.0
3.300	361.6	571.9	326.8	196.3	140.3	78.2	57.3
3.400	322.0	525.2	300.1	181.2	129.8	72.5	53.2
3.500	278.5	483.9	276.5	167.8	120.4	67.3	49.5
3.600	248.9	447.2	255.6	155.7	112.0	62.7	46.1
3.700	227.2	414.6	236.9	145.0	104.4	58.6	43.1
3.800	203.4	385.3	220.2	135.2	97.6	54.9	40.4
3.900	184.4	359.0	205.2	126.5	91.4	51.5	37.9
4.000	171.1	335.3	191.6	118.5	85.8	48.4	35.7
4.100	157.6	313.9	179.4	111.3	80.7	45.5	33.6
4.200	144.6	294.5	168.3	104.7	76.0	43.0	31.7
4.300	134.9	276.8	158.2	98.7	71.7	40.6	30.0
4.400	127.0	260.7	148.9	93.2	67.8	38.4	28.4
4.500	117.2	245.9	140.5	88.1	64.2	36.4	26.9
4.600	107.1	232.4	132.8	83.5	60.9	34.6	25.6
4.700	100.0	219.9	125.7	79.2	57.8	32.9	24.3
4.800	94.1	208.5	119.1	75.2	55.0	31.3	23.2
4.900	88.4	197.9	113.1	71.6	52.3	29.8	22.1
5.000	83.6	188.1	107.5	68.2	49.9	28.4	21.1

Espectros de peligro uniforme por familia y época

	Total				Subducción				Profundidad intermedia			
Periodo	ACT	EFI	EF2	EF3	ACT	EF1	EF2	EF3	ACT	EF1	EF2	EF3
0.010	451.9	414.5	386.4	412.6	366.5	329.6	286.6	293.7	288.3	308.9	315.2	352.6
0.100	449.2	414.8	388.6	416.5	365.1	329.5	286.7	294.3	285.6	310.5	319.0	357.8

0.200	502.8	495.1	511.5	536.4	367.1	336.2	298.9	306.2	373.7	444.1	488.9	515.2
0.300	522.9	501.4	462.1	509.1	391.0	359.9	316.6	332.7	379.5	432.0	412.2	468.6
0.400	604.2	611.2	569.6	576.2	479.4	460.9	414.3	415.5	386.8	486.2	470.0	492.2
0.500	708.9	670.4	723.2	768.9	544.6	489.5	504.3	537.6	488.6	564.3	625.7	671.9
0.600	687.7	725.2	764.7	789.1	533.1	541.2	559.8	567.8	475.6	594.8	635.7	664.7
0.700	808.7	879.2	842.9	738.0	642.7	675.1	637.8	566.8	519.3	682.3	667.6	576.6
0.800	858.9	803.3	682.9	648.6	687.8	638.6	538.2	506.7	549.7	609.1	522.6	499.4
0.900	847.1	728.9	692.3	759.1	707.4	612.6	569.1	606.4	493.6	488.4	485.3	563.7
1.000	772.1	698.3	804.1	984.5	642.6	575.4	645.3	756.3	459.3	481.0	581.5	758.9
1.100	723.4	787.1	987.9	1205.2	613.7	642.9	796.5	934.4	406.6	541.1	693.7	914.3
1.200	803.8	913.9	1244.0	1313.0	688.2	764.6	1029.7	1083.7	412.8	590.2	827.5	877.6
1.300	917.6	1134.7	1420.9	1612.2	785.0	952.8	1175.6	1340.3	481.8	727.2	938.4	1062.2
1.400	1105.3	1354.3	1626.6	1875.5	958.3	1145.7	1375.7	1571.3	558.3	841.8	1027.1	1197.7
1.500	1364.4	1453.2	1660.3	1727.0	1178.3	1238.0	1411.4	1455.4	677.1	881.1	1027.2	1087.0
1.600	1480.1	1470.0	1373.2	1335.1	1328.9	1319.7	1206.2	1160.4	629.3	759.7	770.9	761.6
1.700	1639.0	1591.2	1400.9	1107.5	1512.8	1470.6	1273.4	998.9	614.5	714.6	662.2	561.4
1.800	1843.1	1792.5	1457.4	958.8	1707.1	1666.3	1341.0	869.9	644.2	741.1	644.8	450.1
1.900	2072.5	1881.2	1279.8	808.8	1934.7	1762.1	1186.0	743.6	686.3	728.6	531.0	357.1
2.000	2165.6	1754.2	1047.6	664.3	2034.6	1654.3	982.0	613.7	713.3	653.2	417.5	282.8
2.100	2087.0	1398.0	832.6	525.5	1917.3	1298.3	768.3	481.4	729.9	556.4	350.8	235.4
2.200	1826.8	1093.2	682.9	441.0	1648.7	1007.6	626.7	399.3	687.3	465.8	302.8	204.5
2.300	1475.4	892.2	558.6	380.1	1340.9	815.6	509.0	343.9	537.3	385.4	249.0	176.7
2.400	1189.1	754.2	471.3	341.8	1087.9	696.8	435.1	313.0	416.4	315.5	203.5	152.3
2.500	990.7	647.0	410.6	314.5	919.7	601.2	379.1	292.0	336.0	258.3	169.5	133.7
2.600	826.2	535.5	357.9	285.4	766.6	498.9	332.4	263.9	266.8	212.7	146.1	119.1
2.700	731.4	470.8	333.3	272.4	688.4	444.1	313.4	254.4	226.3	176.6	127.7	105.9
2.800	629.2	417.0	309.2	256.7	590.6	391.6	292.1	240.6	196.3	153.7	115.7	98.1
2.900	557.8	384.1	297.2	251.5	526.3	363.1	282.5	237.2	163.3	135.8	105.3	91.4
3.000	491.9	354.8	283.2	242.7	465.0	335.7	269.3	229.0	145.9	126.0	100.0	88.2
3.100	452.4	339.4	277.4	239.8	429.5	322.3	264.5	227.3	127.8	116.8	94.5	83.5
3.200	401.3	307.8	255.9	218.2	379.0	289.6	240.7	204.4	121.0	113.6	93.6	82.3
3.300	361.6	283.7	241.1	201.3	338.4	267.2	226.0	188.7	111.4	105.2	88.3	76.7
3.400	322.0	258.0	220.5	182.6	300.2	242.5	205.3	170.7	104.8	99.1	85.5	72.2
3.500	278.5	224.8	191.6	158.2	256.1	205.5	174.8	143.8	102.3	97.0	85.2	70.5
3.600	248.9	202.2	172.4	141.2	229.7	185.3	158.0	128.5	93.6	87.3	76.1	63.2
3.700	227.2	185.4	157.5	128.6	210.7	170.5	145.6	117.6	80.8	79.3	67.7	56.8
3.800	203.4	166.6	139.7	115.0	185.1	151.3	126.2	103.8	75.1	77.0	64.6	55.2
3.900	184.4	152.3	125.5	107.3	166.7	137.6	112.6	95.8	70.4	72.3	60.1	53.0
4.000	171.1	142.4	116.0	103.5	155.0	129.7	104.6	93.1	64.5	65.7	54.4	49.2
4.100	157.6	130.8	106.0	96.2	141.3	117.6	94.8	85.3	61.6	62.7	52.1	48.9
4.200	144.6	118.7	96.3	87.7	126.7	104.0	84.0	74.9	61.7	62.3	53.1	50.7
4.300	134.9	109.6	89.4	82.0	117.5	94.9	77.5	68.6	58.6	59.4	51.6	49.4
4.400	127.0	102.6	84.4	78.2	112.1	89.8	74.0	66.5	52.9	53.6	47.9	45.0
4.500	117.2	94.3	79.1	72.9	103.9	82.7	68.4	61.7	49.6	49.5	46.1	42.6
4.600	107.1	86.0	73.1	66.8	93.3	74.0	61.2	54.2	47.6	47.9	47.0	43.3
4.700	100.0	80.4	69.2	62.7	85.0	67.7	55.5	48.4	47.2	48.1	48.9	44.8
4.800	94.1	75.9	66.3	59.4	79.0	63.3	51.9	45.0	45.8	47.4	48.4	44.1
4.900	88.4	71.7	63.4	56.6	74.9	59.9	49.9	43.3	42.6	45.0	45.6	40.9
5.000	83.6	68.3	61.3	54.3	71.2	57.1	47.9	42.0	39.0	43.7	43.6	38.6